

PIC Group 03 - Fittipaldi

Formula E-Vee

Mid-Term Report

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April 2, 2026

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List of Symbols and Acronyms

This section defines the key symbols and acronyms used throughout this document.

Acronyms

AHP	Analytic Hierarchy Process
Al	Aluminum
BMS	Battery Management System
BoP	Balance of Plant
CAD	Computer-Aided Design
COTS	Commercial Off-The-Shelf
CO₂	Carbon Dioxide
DC	Direct Current
DOT	Design Option Tree
DoD	Depth of Discharge
EESM	Externally Excited Synchronous Motor
EMI	Electromagnetic Interference
ESG	Environmental, Social, and Governance
ESU	Energy Storage Unit
EU	European Union
EV	Electric Vehicle
FC	Fuel Cell
FEA	Finite Element Analysis
FIA	Fédération Internationale de l'Automobile
GHG	Greenhouse Gas
GWP	Global Warming Potential
H₂	Hydrogen
HV	High-Voltage
ICE	Internal Combustion Engine
IMD	Insulation Monitoring Device
IGBT	Insulated-Gate Bipolar Transistor
ILCD	Int. Reference Life Cycle Data System
IST	Instituto Superior Técnico
LCA	Life Cycle Assessment
LFP	Lithium Iron Phosphate
MTBF	Mean Time Between Failures
NACA	National Advisory Committee for Aeronautics
NdFeB	Neodymium-Iron-Boron
NMC	Nickel Manganese Cobalt
OAT	Organic Additive Technology
OEM	Original Equipment Manufacturer
PCM	Phase Change Material
PEM	Proton Exchange Membrane
PM	Particulate Matter
PMSM	Permanent Magnet Synchronous Motor
RMI	Responsible Minerals Initiative
SDG	Sustainable Development Goals
SiC	Silicon Carbide
VCU	Vehicle Control Unit
VW	Volkswagen

Symbols

A	Frontal area [m ²]
<i>a_x</i>	Longitudinal acceleration [m/s ²]
<i>a_y</i>	Lateral acceleration [m/s ²]
Ah	Capacity (Ampere-hour)
<i>C_d</i>	Drag coefficient [-]
<i>C_l</i>	Lift coefficient [-]
<i>C_{rr}</i>	Rolling resistance coefficient [-]
<i>E</i>	Energy [kWh]
<i>E_{model}</i>	Modeled energy consumption
<i>E_{total}</i>	Total required energy
<i>F_{down}</i>	Aerodynamic downforce [N]
<i>g</i>	Acceleration of gravity [m/s ²]
<i>I</i>	Current
<i>I_{ch}</i>	Charge current
<i>I_{dis}</i>	Discharge current
km	Kilometers
kW	Kilowatt
L	Volume (liters)
<i>m</i>	Mass [kg]
max	Maximum value operator
<i>n_p</i>	Number of parallels
<i>n_s</i>	Number of series
<i>P</i>	Power [kW]
<i>P_{bat}</i>	Battery power [W]
<i>P_{loss}</i>	Power losses
<i>P_{wheel}</i>	Power at the wheels [W]
<i>R</i>	Electrical resistance
SoC	State of Charge
SoH	State of Health
<i>T</i>	Torque [Nm]
<i>T_{ch}</i>	Maximum charging temperature
<i>T_{max}</i>	Maximum temperature
<i>v</i>	Vehicle speed [m/s]
V	Voltage
<i>V_{min}</i>	Minimum voltage
<i>V_{nom}</i>	Nominal voltage
W	Power (Watts)
Wh/L	Volumetric density
<i>γ_{brake}</i>	Regen. braking distribution factor [-]
<i>η_{drive}</i>	Drivetrain efficiency [-]
<i>η_{reg}</i>	Regenerative efficiency [-]
<i>κ</i>	Track curvature [1/m]
<i>ρ</i>	Air density [kg/m ³]

1 Introduction

The legacy of Formula Vee has always been defined by accessibility, providing a platform where engineering ingenuity outweighs financial expenditure. As the automotive industry undergoes a paradigm shift toward electrification, the Formula E-Vee project arises not merely as a technical exercise in powertrain conversion, but as a commitment to the future of sustainable grassroots motorsport. Developed by PIC Group 03 – Fittipaldi, a group of Mechanical Engineering students from Instituto Superior Técnico, this project seeks to reconcile the raw, competitive spirit of the original Formula Vee with the urgent demands of a zero-emission era.

The project's vision is anchored in the belief that high-performance engineering must coexist with environmental stewardship. By integrating principles of the circular economy with advanced engineering solutions, the objective is to deliver a vehicle that is both cost-effective for novice drivers and technically sophisticated for the modern track. This Mid-Term Report documents the foundational phase of this endeavor. It begins by establishing a comprehensive benchmarking of existing racing platforms and defining the sustainable strategy that governs the overarching design choices.

The core of this work lies in the exploration of four distinct vehicle philosophies: the Sustainable concept, focused on minimizing the environmental footprint through circular economy principles and maximum component reuse; the Economical concept, prioritizing budget accessibility and straightforward integration using mature industry standards; the Performance concept, centered on maximizing track capabilities through highly integrated, lightweight, and high-yield solutions; and the Innovative concept, pushing the boundaries of modern automotive engineering by adopting state-of-the-art technologies. Through a rigorous market analysis and a multi-criteria trade-off evaluation, this report navigates the complexities of component selection across the vehicle, providing the technical roadmap necessary to transition from conceptual theory to a functional, competitive, and truly green racing prototype.

2 Market Analysis

Formula Vee is a low-cost single-seater defined by simple design and low mass [4]. In parallel, electric motorsport has demonstrated high performance with zero operational emissions, as seen in Formula E [6].

2.1 Benchmarking of Racing Platforms

Formula Vee cars produce 30 kW and weigh 480 kg, reflecting simple and low-cost design [4].

Electric racing karts deliver 20–30 kW with 120–160 kg and 4–8 kWh batteries, suited for short sessions [13]. Formula Student EV vehicles reach 80 kW and 250 kg, while Formula E operates at 250 kW with 54 kWh [7, 6].

Hydrogen racing achieves high power but with significantly higher cost and system complexity [9].

Based on these benchmarks, the Formula E-Vee target was defined as 50 kW, 400 kg, and 13 kWh (Table 1).

Category	Power	Weight	Energy	Top Speed	Cost
Formula Vee	30 kW	480 kg	–	180 km/h	5k–55k EUR
Racing Electric Kart	20–30 kW	120–160 kg	4–8 kWh	120–140 km/h	15k–25k EUR
Formula Student EV	80 kW	250 kg	6–7 kWh	120 km/h	60k–100k EUR
Hydrogen Racing	200 kW	1100 kg	Fuel cell	210 km/h	300k–800k EUR
Formula E	250 kW	900 kg	54 kWh	320 km/h	>1M EUR
Formula E-Vee	50 kW	400 kg	13 kWh	170 km/h	<30k EUR

Table 1: Benchmark of Racing Platforms

To position the proposed vehicle within the motorsport landscape, cost–power and power-to-weight relationships are shown in Figure 1.

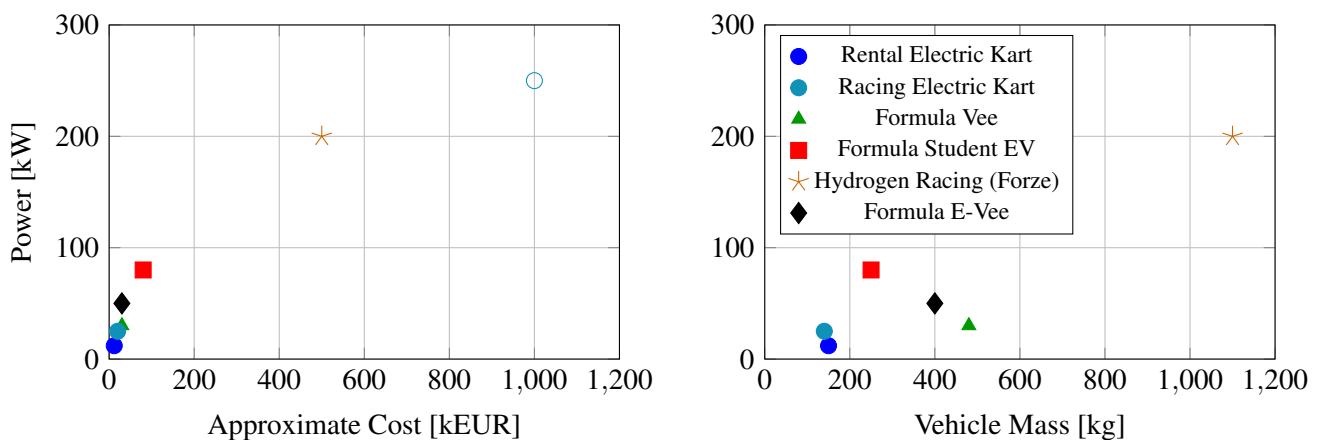


Figure 1: Comparison of Representative Racing Categories and Electric Track Vehicles

2.2 Design Principles from Formula Vee

Formula Vee vehicles follow design principles that remain applicable to the Formula E-Vee despite the electric drivetrain [5, 4].

Steel tubular space frames offer a balance between stiffness, low cost, and manufacturability [5]. Low mass is critical for performance and efficiency, particularly given battery weight.

Mechanical simplicity improves reliability and reduces maintenance, supporting the avoidance of unnecessary complexity.

Standardized components reduce development risk and facilitate maintenance, favoring commercially available electric solutions.

The observations in Table 2 summarize these principles, highlighting their relevance for cost-effective and reliable electric vehicle design.

Design Aspect	Observation	Relevance for Formula E-Vee
Chassis architecture	Steel tubular space frame	Low cost and easy fabrication
Vehicle mass	Very lightweight design	Improves efficiency and performance
Mechanical simplicity	Minimal complex systems	Improves reliability
Standardized components	Use of proven parts	Reduces development risk
Compact packaging	Efficient layout of components	Supports EV integration
Ease of maintenance	Accessible components	Important for racing operation

Table 2: Design Characteristics Observed in Competitive Formula Vee Vehicles

3 Sustainable Development Strategy

The lifetime of the Formula E-Vee Fittipaldi comprises the vehicle production, track operation (minimum 100 km/day $\times$ 10 years = 50 000 km), as said in the Project Guide and end-of-life treatment of all components.

The methodology used will be Streamlined Life Cycle Assessment (LCA), following the Sustainability book referenced in the Project Guide [44]. Rough values of the Global Warming Potential (GWP) and CO_2 will be used.

The assessment will be done with a philosophy of cradle-to-grave, covering phases of production and manufacturing, use, and end of life. Transport and logistics are excluded from this screening assessment.

3.1 Powertrain

The main focus of the powertrain will be to choose components with low chemical toxicity and mining geological impact while having a high useful life.

ESU (Energy Storage Unit) - The Battery Pack will be the component with the most environmental impact, having a very high CO_2 Kg per kWh of battery and medium recyclability. Cobalt Cells must be studied very carefully because they often have a high Environmental, Social, and Governance (ESG) impact, given the fact that cobalt mining frequently has a negative social impact on the local community. The possibility of Green Hydrogen concepts is a very sustainable choice with a very low ESG impact and a more reasonable GWP when compared with other options.

Motor - When looking at the sustainability impact of electric motor, the main focus is on the permanent magnets who usually contain ≈ 0.8 kg Neodymium-Iron-Boron (NdFeB) per unit. NdFeB is a rare-earth mineral that is obtained through high ecological and hydrological impact mining, while having high CO_2 and $<1\%$ recovery rate at a rare-earth recovery facility. Motor concepts without permanent magnets must be investigated.

Cooling - Cooling efficiency must be optimized and maximized to prevent premature wear of the ESS, so it has the maximum useful life possible. With the possible use of biodegradable coolant, to maximise end-of-life recyclability.

3.2 Structures

This is where to optimise the re-use of parts and support the circular economy. The parts must be selected and designed to achieve a long useful life.

Chassi - The strategy for the chassis will be to have as close to a 100% recyclable chassis, to maximize end of life sustainability. The possibility of re-modifying a Beetle chassis must be studied because it takes advantage of the circular economy.

Suspension/Steering - When it comes to suspension and steering, the key is simplicity; fewer components result in less environmental impact.

Braking - For the braking system there are two options: Drum brakes or Disc brakes. Drum Brakes have fewer particle emissions, making them the best sustainable option, meaning they must be prioritised in the concept design.

3.3 Sustainable Development Goals (SDGs)

Table 3 quantifies the impact that the powertrain and chassis have on the project's compliance with the SDGs that were assessed as directly applicable to this project scope.

Icon	#	SDG	PT	ST	Justification
	3	Good Health & Well-being	—	—	Nickel-Manganese-Cobalt (NMC) and carbon fibre present human toxicity and labour risks. Lithium-Iron-Phosphate (LFP) batteries, a magnet-free motor and a fuel cell are possibilities for a more sustainable powertrain.
	4	Quality Education	+	+	The development of a zero-emission vehicle within an academic setting provides hands-on engineering education aligned with sustainability principles.
	7	Affordable & Clean Energy	++	○	Zero direct Greenhouse Gas (GHG) emissions in operation.
	9	Industry & Innovation	+	+	Circular economy motor, Al-Air fuel cell, Phase Change Material (PCM)/heat pipe cooling, and Beetle chassis retrofit.
	11	Sustainable Cities	+	+	Noise <90 dB met by all concepts. Drum brakes emit ≈0.5 mg/km PM ₁₀ vs 4–12 mg/km for discs.
	12	Responsible Consumption	+	+	LCA used as design tool; biodegradable coolant; salvaged motor; Beetle retrofit; VW Type 1 parts reuse.
	13	Climate Action	++	+	Reduction in operational CO ₂ . Steel chassis and suspension keep embodied carbon low.
	14	Life Below Water	+	+	Zero GHG emission and pollutant emissions reduce acid rain and contamination of aquatic ecosystems. By not using fossil fuels reduces the need of oil station in the ocean.
	15	Life on Land	+	+	Electric operation produces no exhaust pollutants and noise levels <90 dB, minimising habitat disturbance and acoustic stress on terrestrial fauna.
	17	Partnerships	+	○	IST–industry academic partnership; RMI-certified sourcing; EU Battery Regulation 2023 traceability.

++ strong positive + positive ○ neutral — negative.

Table 3: SDG Impact Assessment

4 Risk Analysis

4.1 Introduction and Methodology

The technical risk assessment identifies and ranks factors threatening the Formula E-Vee project's overall viability, including its technical performance, sustainability, internal management, and cost. Hazards are quantified on a **Risk Map** plotting **Probability** (*Rare to Certain*) against **Impact** (*Negligible to Catastrophic*) to generate a prioritized **Risk Score**. High-ranking risks receive targeted mitigation strategies to reduce their likelihood or severity, after which a post-mitigation map is plotted to visualize the residual risks.

4.2 Technical Risk Register

The following technical risks have been identified, assessed, and mapped. Each entry details the specific hazard, the planned mitigation strategy, and the resulting transition in the risk score, summarized in Table 4.

ID & Categ.	Hazard Description	Mitigation Strategy	Risk Score (Pre → Post)
R01a Energy (H2)	High-pressure hydrogen tank rupture or leak during a crash or track operation, leading to rapid gas expansion or fire.	Use certified tanks meeting regulatory safety factors. Integrate pressure release and H2 sensors to disconnect HV below lower flammability limits.	High → Medium (Possible/Cat. → Rare/Cat.)
R01b Energy (HV)	Battery thermal runaway or insulation breakdown during the demanded 30-minute fast charge or 50 kW track discharge.	Implement active liquid cooling. Set strict BMS thermal cut-offs according to cell datasheets and integrate an FIA-compliant IMD.	High → Medium (Possible/Cat. → Rare/Cat.)
R02 Chassis	Structural failure during operation or when retrofitting the spaceframe with new EV components.	Validate torsional stiffness and load paths via FEA to withstand the target 1.6 G lateral and 1.2 G longitudinal peak dynamic loads.	High → Medium (Possible/Cat. → Unlikely/Cat.)
R03 Finance	Budget overruns driven by the supply chain or design options/rework breaching the strict 30,000€ limit (15,000€ power, 5,000€ chassis).	Strictly adhere to the established baseline cost plan through continuous budget tracking and design-to-cost validation across all subsystems.	High → Low (Likely/Mod. → Unlikely/Mod.)
R04 Performance	Failing the 6-lap pace target (< 2m15s/lap) under the mandatory 50 kW power cap.	Constrain vehicle mass strictly within [385, 425] kg and optimize aero/rolling resistance for >80% efficiency.	High → Low (Possible/Maj. → Unlikely/Mod.)
R05 Maintenance	Exceeding the 1.0% / 100 km maintenance cost cap due to complex repairs.	Implement modular sub-assemblies and maximize the use of standardized components to minimize part costs.	High → Very Low (Likely/Mod. → Rare/Mod.)
R06 Mechanical	Instantaneous 200 Nm electric torque exceeding the mechanical limits of the drivetrain, and both the new and legacy Beetle components.	Implement VCU software for torque output, and structurally reinforce mounts to withstand the simulated stresses of the 200 Nm peak torque.	Critical → Medium (Likely/Cat. → Unlikely/Maj.)
R07 Daily Usage	Failing the 100 km/day demand due to insufficient battery capacity for consecutive sessions.	Size battery pack with a 20% energy safety buffer above 30 km baseline.	Medium → Low (Possible/Mod. → Rare/Maj.)
R08 Environmental	Inadequate end-of-life battery disposal and excess manufacturing waste violating UN's responsible production goal and LCA targets.	Conduct a full LCA during design and establish strict supplier recycling agreements.	Medium → Very Low (Possible/Mod. → Unlikely/Min.)
R09 Safety	Safety hazards (collisions, rollovers, fires, and HV/LV faults) compromising operational safety.	Validate chassis via FEA for FIA /Formula Vee compliance and integrate IMD/BMS systems for automatic high-voltage shutdown upon fault.	Critical → Medium (Likely/Cat. → Unlikely/Cat.)
R10 Durability	Failing 10-year / 50,000 km lifetime requirement due to battery degradation or component fatigue.	Design components according to their expected operating life/conditions, monitor battery SoH, schedule periodic inspections, and apply anti-corrosion treatments.	High → Very Low (Possible/Maj. → Unlikely/Min.)
Continued on next page...			

R11 Man- agement	Internal communication breakdowns and uneven workload distribution leading to subsystem integration conflicts and schedule delays.	Maintain established documentation standards on shared platforms, implement collaborative task pairing and conduct regular meetings.	Medium → Very Low (Likely/Min. → Rare/Min.)
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Table 4: Technical Risk Register Detailing Pre- and Post-Mitigation Risk Evaluations, including integrated ESG Constraints.

4.3 Pre and Post-Mitigation Risk Maps

The transition of prioritized risks - before and after the implementation of the engineering strategies detailed in Table 4 - is visualized in Figure 2. The map’s color gradient categorizes the risk score previously detailed, scaling continuously from Red (Critical) through Orange (High) and Yellow (Medium), down to Light and Dark Green (Low to Very Low).

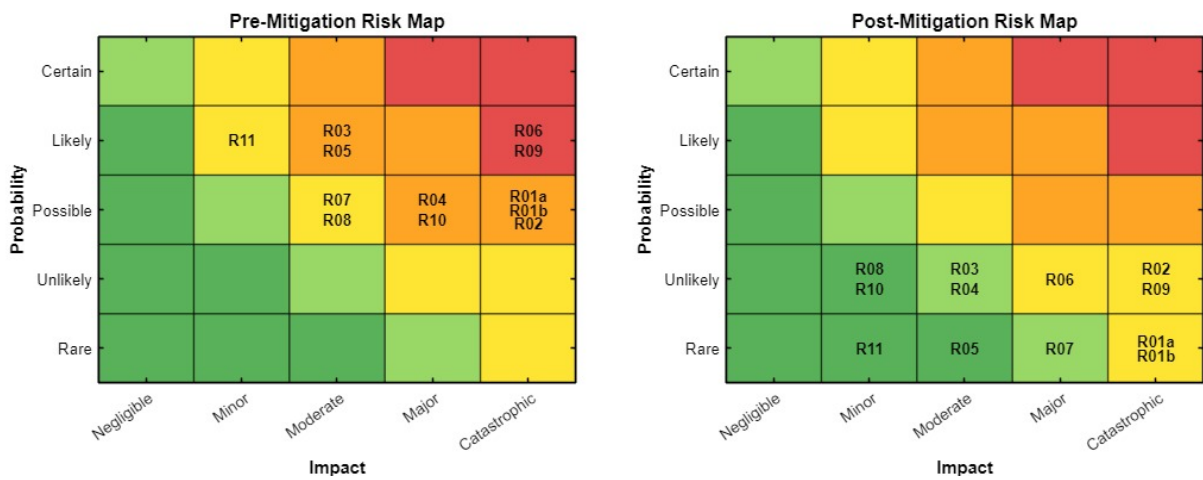


Figure 2: Pre- and Post-Mitigation Technical Risk Maps.

4.4 Conclusion

The initial technical risk assessment identified high-voltage safety, the mechanical limits of the components, and powertrain instability as the most critical threats. Furthermore, operational and team management risks, like task delays, cross-department conflicts, and decision-making bottlenecks, were recognized for their potential to severely disrupt the project’s viability. Left unmitigated, these vulnerabilities could induce catastrophic structural or electrical failures, directly jeopardizing driver safety, track performance, and financial constraints.

To counter these threats, the team implemented mitigation strategies such as validating structural integrity via Finite Element Analysis (FEA), implementing torque output control, integrating BMS/IMD safety mechanisms, and enforcing constant internal communication. As illustrated by the post-mitigation Risk Map, this approach successfully downgrades all Critical and High risks to acceptable levels (capped at Medium), ensuring the Formula E-Vee remains a highly reliable, cost-effective, and safe platform for sustainable racing.

5 Concepts Introduction

The primary objective of this project is the development of a Formula Vee vehicle designed with a commitment to zero carbon emissions and the complete elimination of dependence on fossil fuels. To guide the research and development process, a strategic framework was established based on four fundamental concepts:

Sustainability Economic Performance Innovation

These pillars were specifically defined due to the inherent challenge of balancing them within an engineering context. Consequently, each subsystem was required to propose a distinct solution for each of these pillars, except in specific cases where the conceptual influence was considered insignificant.

To ensure an objective selection process, we implemented a standardized evaluation methodology using criteria such as cost, efficiency, sustainability, complexity, performance, and reliability. Each criterion was linked to a high-level requirement and assigned a specific weight according to its strategic relevance. Furthermore, as each subsystem serves a unique technical purpose, specific metrics were defined to evaluate its concepts. This approach ensured that both high-level and low-level requirements were rigorously met, making the classification and clarification of design concepts significantly more efficient and transparent.

6 Estoril Circuit Lap Simulator

Before diving into the detailed evaluation of individual components, it is crucial to establish the dynamic boundary conditions and performance targets for the vehicle. To achieve this, the team developed a custom point-mass lap simulator focused on the Circuito do Estoril. By concentrating the vehicle's mass at its center of gravity, the model simplifies multi-body interactions into a particle dynamics problem.

The core logic relies on an iterative spatial integration over a discretized track to define the vehicle's absolute physical limits.

6.1 Track Geometry and Continuity

To ensure a stable numerical derivative, the track coordinates are processed into a differentiable path. This continuity is vital because the local curvature (κ) dictates the instantaneous lateral force demand:

$$\kappa(s) = \frac{|x'y'' - y'x''|}{(x'^2 + y'^2)^{3/2}} \quad (1)$$

Consistent curvature calculation prevents non-physical acceleration "spikes" and ensures that the simulated racing line reflects a realistic trajectory where lateral load changes are smooth.

6.2 Dynamic Limit Envelopes (Three-Pass Logic)

The velocity profile is determined by solving for the minimum of three constraint envelopes, as illustrated in Figure 3. The cornering limit (v_{curve}) represents the steady-state equilibrium where tire grip, aided by velocity-dependent downforce, balances centrifugal force:

$$v_{curve}(s) = \sqrt{\frac{a_{y,max} \cdot \left(1 + \frac{0.5\rho C_L A v^2}{mg}\right)}{\kappa(s)}} \quad (2)$$

Kinematic updates are performed spatially using Equation 3, allowing the model to "look ahead" for braking points (Backward Pass) and "push" for acceleration (Forward Pass):

$$v_{i+1} = \sqrt{v_i^2 + 2a_x \Delta s} \quad (3)$$

The available acceleration (a_x) is governed by the **friction circle**, which assumes that the tire's total capacity is a shared resource between longitudinal and lateral demands:

$$a_{x,avail} = a_{x,max} \sqrt{1 - \left(\frac{a_y}{a_{y,max}}\right)^2} \quad (4)$$

6.3 Electromechanical Power and Regeneration

Power at the wheels (P_{wheel}) is modeled by summing the rate of work required to overcome inertia and environmental losses (aerodynamic drag and rolling resistance):

$$P_{wheel} = \left(m \cdot a_x + \frac{1}{2} \rho C_d A v^2 + C_{rr} (mg + F_{down}) \right) \cdot v \quad (5)$$

The battery power (P_{bat}) accounts for directional powertrain losses. Efficiency is inverted between traction and regeneration to reflect the physics of energy flow:

$$P_{bat} = \begin{cases} P_{wheel} / \eta_{drive} & \text{if } P_{wheel} \geq 0 \\ P_{wheel} \cdot \eta_{reg} \cdot \gamma_{brake} & \text{if } P_{wheel} < 0 \end{cases} \quad (6)$$

The factor γ_{brake} justifies the regenerative strategy, representing the torque fraction allocated to the rear motor while respecting the forward load transfer that reduces rear-axle grip during braking.

The culmination of these dynamic limits is visualized in Figure 3, which maps the simulated velocity profile over the Estoril circuit. This trace highlights the aggressive acceleration zones and heavy braking events inherent to the track's layout, directly dictating the peak power demands and thermal loads that the powertrain and braking systems must endure.

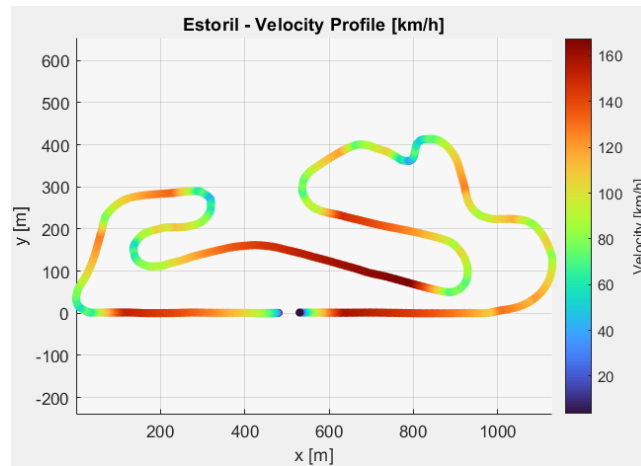


Figure 3: Velocity Profile

6.4 Strategy and Results

The simulation evaluates a 6-lap stint. To ensure reliability, **Hotlaps** operate at 100% performance, whereas **Outlaps** and **Inlaps** are scaled down to 85% and 70% respectively. This differentiation ensures the Net Energy calculation ($E_{used} - E_{regen}$) remains conservative for battery safety.

7 Subsystem Concept Evaluation

Guided by the Risk Map, Sustainability Strategy, and initial Market Analysis, this section presents an in-depth technical evaluation by each subsystem department.

7.1 Powertrain

7.1.1 Inverters

Market Overview

To ensure the selection of the optimal inverter for the Formula E-Vee powertrain, a preliminary market analysis was conducted. The market currently offers several distinct inverter typologies, each aligning appropriately with the project's overarching concepts:

- **Sustainable Concept (Circular Economy & DIY Typology):** A truly sustainable concept prioritizes the entire lifecycle of the component. The commercial inverter market implies a significant carbon footprint due to rare-earth mining and silicon doping. Therefore, this approach points towards open-source architectures (e.g., Lebowski projects) or completely custom in-house builds using scavenged and repurposed electronic components, strictly applying circular economy principles.
- **Economical Concept (Standard COTS Si-IGBT):** For a cost-effective racing concept that remains accessible to novice teams—adhering to the original spirit of Formula Vee—standard Silicon Insulated-Gate Bipolar Transistors (Si-IGBT) are the industry standard. This technology is highly massified, thoroughly reliable, and relatively inexpensive (e.g., Unitek Bamocar series, Sevcon Gen4, Curtis Controllers).
- **Performance Concept (Integrated Powertrain Typology):** In high-level motorsport, packaging, weight distribution, and loss reduction are crucial. The market trend for maximizing pure performance is the utilization of integrated systems, where the motor and inverter share a single housing or are perfectly paired from the factory (e.g., AMK drives, McLaren Applied). This minimizes HV wiring and optimizes dynamic response.
- **Innovative Concept (Silicon Carbide - SiC Technology):** Regarding the vanguard of commercial power electronics, Silicon Carbide (SiC) transistors dominate. They allow for exceptionally high switching frequencies and operate at higher temperatures, drastically reducing thermal losses. Brands like Cascadia Motion and Rinehart lead this segment of high-power-density components.

Selection and Technical Justification

Based on the aforementioned market analysis, specific inverter solutions were selected to represent each core vehicle concept. The technical justification for each choice versus other market alternatives is detailed below.

Sustainable Concept: In-house Built Inverter (Repurposed)

The choice to build an inverter in-house using repurposed components was made because no commercial off-the-shelf (COTS) option can compete with the near-zero manufacturing carbon footprint of reusing existing electronics [14]. Rather than purchasing a new "green" inverter, which would still require global transport and raw material extraction, the team opted for pure circular economy, sacrificing integration ease for maximum sustainability.

By relying heavily on salvaged and repurposed components, this in-house solution presents an extremely low **cost** and exceptional **sustainability**, actively promoting electronic waste reduction. However, matching disparate parts inherently sacrifices optimal **efficiency** compared to modern commercial boards. The development process introduces significant **complexity**, requiring custom printed circuit board design and from-scratch control programming. Consequently, the overall **performance** remains limited by the weakest component in the loop, while the unknown prior thermal and mechanical fatigue of the reused parts raises inevitable concerns regarding the long-term **reliability** of the powertrain.

Economical Concept: Bamocar D3

Among economical market options, the Bamocar D3 stands out and was selected due to its user-friendly software ecosystem and proven robustness in academic motorsport environments (such as Formula Student). While other budget-friendly inverters require complex proprietary tools, the Bamocar offers a smooth learning curve and excellent documentation[15], saving critical team development time.

The Bamocar D3 serves as a highly accessible unit that perfectly balances the **cost** constraints of entry-level racing budgets. Utilizing mature Si-IGBT technology, it delivers adequate **efficiency** and sufficient track **performance** to ensure class competitiveness without focusing on extreme power delivery. From an environmental standpoint, its **sustainability** is considered intermediate, as it relies on standard industrial manufacturing processes. One of its greatest advantages is the very low integration **complexity**, offering exceptionally straight-

forward physical mounting and software tuning. Above all, its extensive application in light electric vehicles and student motorsport guarantees outstanding **reliability**.

Performance Concept: AMK Integrated

Rather than selecting an isolated high-power inverter, the AMK integrated architecture was chosen. In the performance market, separating the motor and inverter implies longer HV cables, added weight, and increased Electromagnetic Interference (EMI) challenges [17]. The AMK solution resolves this by perfectly integrating both systems, delivering a dense powertrain focused on instantaneous torque response and superior chassis packaging.

Reflecting its precision engineering and high-yield focus, the AMK integrated architecture commands a higher financial **cost** but compensates with outstanding track capabilities. The factory pairing of the motor and inverter yields very good electrical **efficiency** and exceptional dynamic **performance**, offering high athletic capacity and rapid transient response. By suppressing the traditional high-voltage wiring loom, the system gains a moderate advantage in **sustainability** through raw material savings. While the unified packaging simplifies physical integration, the advanced telemetry demands introduce a moderate level of engineering **complexity**. Ultimately, the system provides very robust **reliability**, having been specifically designed to withstand severe racing vibrations and thermal loads.

Innovative Concept: Cascadia SiC

The Cascadia unit was selected as it represents technologically viable innovation ready for track deployment. While more exotic laboratory technologies exist, Cascadia Motion dominates the commercial market for compact SiC inverters. This technology drastically minimizes switching losses compared to standard IGBTs, offering unmatched power density [19].

Utilizing latest-generation silicon carbide semiconductors, the Cascadia unit demands a very high **cost** but pushes the boundaries of power electronics. This technology provides exceptional **efficiency** by drastically minimizing switching losses, which in turn enhances its operational **sustainability** by reducing the energy wasted as heat and the associated charging emissions. Extracting extreme power levels from a highly reduced volume guarantees maximal **performance**. However, operating at these switching frequencies introduces significant **complexity**, requiring strict electrical noise isolation and advanced control algorithms. Provided the required flawless cooling system is maintained to prevent thermal shock, the unit offers excellent modern **reliability** for high-demand track driving.

Trade-off and Results

To synthesize the qualitative evaluation into a structured multi-criteria decision model, a trade-off matrix was developed (Table 5). The criteria were scored from 1 (Poor / Highly undesirable for the project) to 5 (Excellent / Highly desirable).

It is important to note that factors such as *Cost* and *Complexity* operate inversely; therefore, a high financial cost or a high level of integration difficulty represent negative impacts on the project's feasibility and logically receive lower numerical scores.

Evaluation Criteria	Sustainable	Economical	Performance	Innovative
Cost	5 (1k–3k €)	3 (6k–8k €)	1 (12k–18k €)	2 (10k–15k €)
Efficiency	2 (90–94%)	3 (95–97%)	4 (96–97%)	5 (97–99%)
Sustain.	5 (Component reuse)	3 (Standard materials)	3 (Integrated, less wiring)	3.5 (High power density, SiC)
Complexity (5 = <i>simpler</i>)	2 (Custom, difficult)	5 (Simple, robust)	4 (Moderate, integrated)	2 (High control complexity)
Perform.	2 (Limited)	3 (Adequate)	4 (Good overall)	5 (High power, fast response)
Reliability	2 (Uncertain, reused parts)	5 (Very proven)	5 (Very robust)	4 (Modern, reliable)

Table 5: Inverter Concept Trade-off Matrix

7.1.2 Motors

Market Overview

The current EV market offers several motor topologies with distinct cost, efficiency, weight, and sustainability trade-offs [23]. To align with the proposed vehicle concepts, the following four architectures were explored:

Electrically Excited Synchronous Motors (EESM): Utilizing a wound rotor supplied with DC current instead of permanent magnets (e.g., Renault Zoe), this represents the **Sustainable** concept. Eliminating rare-earth metals severely reduces environmental mining impact. Furthermore, sourcing salvaged Original Equipment Manufacturer (OEM) parts virtually negates the manufacturing carbon footprint, offsetting the fact that these motors are slightly bulkier.

Permanent Magnet Synchronous Motors (PMSM) - Radial Flux: This standard cylindrical geometry where the magnetic field expands outward radially (e.g., Bosch SMG) fits the **Economical** concept. As the mass-market industry standard, they are highly reliable, ubiquitous in salvage yards, and offer a cost-effective, plug-and-play solution with rigid OEM housings.

High-Performance PMSM: These specialized radial or transverse flux motors (e.g., AMK DD) push the boundaries of power density, perfectly fitting the **Performance** concept. Their minimal weight and tiny form factor enable advanced setups like direct in-wheel mounting for active torque vectoring, albeit at a severe financial premium.

PMSM - Axial Flux: Here, magnetic flux travels parallel to the rotation axis, yielding a highly torque-dense “pancake” geometry (e.g., EMRAX). Fitting the **Innovative** concept, it offers immense longitudinal packaging advantages for the narrow tubular spaceframes characteristic of open-seater racing cars.

Selection and Technical Justification

Building on the market analysis, a specific motor was selected to physically represent each of the four design concepts. The following subsections detail the technical justifications, key specifications, and inherent compromises of each choice.

Sustainable Concept: EESM - Renault Zoe R110

The R110 was selected primarily because its EESM topology avoids the use of rare-earth permanent magnets [21]. While highly sustainable and cost-effective ($\sim 1,000$ €, salvaged), the trade-off is significant in weight. At approximately 50 kg, it yields a poor power-to-weight ratio (~ 1 kW/kg), directly impacting vehicle dynamics and track performance. Furthermore, the bulky OEM geometry (shown in Figure 4) makes integration into the narrow Formula Vee chassis highly complex.

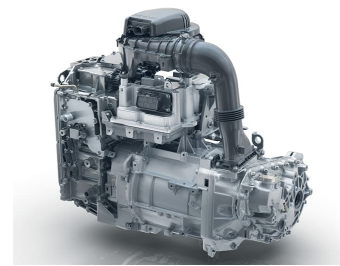


Figure 4: Renault Zoe R110

Economical Concept: PMSM - Bosch SMG 180/120

The Bosch SMG series represents the pinnacle of accessible OEM reliability [20]. Widely available in salvage yards, it ensures the powertrain stays well below budget ($\sim 1,000$ €). It provides better power density than the EESM (~ 30 kg; 1.6 kW/kg), and its rigid OEM housing guarantees excellent track reliability. Its standard cylindrical geometry (Figure 5) is simpler to adapt than the Renault unit. This straightforward shape facilitates custom mounting and standard cooling solutions, significantly reducing the engineering hours required for chassis integration compared to more exotic topologies.

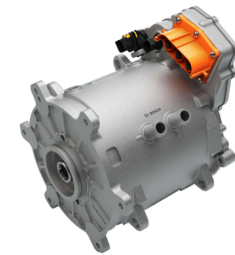


Figure 5: Bosch SMG 180/120

Performance Concept: PMSM - 2x AMK DD5

To maximize on-track performance, a dual-motor setup using AMK DD5 motors was selected to be positioned directly in-wheel at the rear [22]. This setup eliminates the need for a heavy central differential and driveshafts, drastically reducing mechanical losses while enabling active torque vectoring. The performance metrics are exceptional: weighing a combined total of only ~ 7 kg (3.5 kg each), the package offers a massive 7.14 kW/kg power density. However, this architecture presents a severe financial compromise, pushing the powertrain cost to approximately $8,000$ € — this makes it the most expensive option on the table. Their compact profile is shown in Figure 6.



Figure 6: AMK DD5

Innovative Concept: PMSM - Axial Flux - EMRAX 188

The EMRAX 188 was selected as the innovative solution [10]. Priced around $3,000$ € to $4,000$ €, it comfortably aligns with the budget while delivering remarkable performance (~ 8 kg; 6.25 kW/kg) and maintaining high efficiency. The "pancake" geometry is extremely simple to package longitudinally in the chassis. However, axial flux motors have extremely tight internal tolerances (small air gaps between the rotor and stator). Under hard track use, any flex in the chassis or motor mounts could cause catastrophic internal rubbing. Therefore, the primary drawback is the need for a highly rigid, mechanically safe integration, alongside a meticulous cooling loop design to prevent overheating and power loss (Figure 7).



Figure 7: EMRAX 188

Trade-Off and Results

To objectively evaluate these options, the four concepts were assessed using a weighted scoring matrix based on Cost, Efficiency, Sustainability, Complexity, Performance, and Reliability. Higher scores indicate a more favorable outcome (e.g., a Complexity score of 5 means it is very simple to integrate). The results are summarized in Table 6.

Metric	Sustainable	Economical	Performance	Innovative
Cost	5 (~1k, salvaged)	5 (~1k, salvaged)	1 (massive impact, ~8k)	3 (~3/4k, aligns with budget)
Efficiency	2 (~85-89%)	3 (~90-93%)	5 (~95-97%)	4.5 (~94-96%)
Sustain.	5 (motor type + reused)	3 (reused)	2 (2 motors + type of motor)	3 (type of motor)
Complexity	2 (OEM geometry, hard to adapt)	3 (OEM, but simpler geometry)	2 (simple geometry, harder in-wheel integration)	4 (simple geometry, but demands rigid mounting)
Perform.	1 (~50kg; 1kW/kg)	2 (~30kg; 1.6kW/kg)	5 (~7kg total; 7.14kW/kg)	4 (~8kg; 6.25kW/kg)
Reliability	3 (OEM, but motor type brings issues)	5 (OEM and rigid housing)	4 (could suffer from hard track use)	3 (needs highly safe integration)

Table 6: Motor Concept Trade-off Matrix

7.1.3 Energy Storage Unit (ESU)

Market Overview

Technical Benchmarking

To position this prototype, we split the market into three tiers based on primary driver, longevity and autonomy recovery, as summarized in Table 7.

Feature	Formula Vee (ICE)	Formula E (Pro EV)	Formula Student	Hydrogen Racing	Formula E Vee Prototype
Primary Driver	Low cost	Power density	Perf./weight	Range / demo	Balance
Longevity	Re-used	Seasonal	Seasonal	FC + H ₂ safety	1667 cycles
Charging	Gas refill	Dedicated charger	Dedicated charger	H ₂ refill	400V DC fast

Table 7: ESU Technical Benchmarking Across Motorsport Tiers

As a prototype, this design demonstrates that an electric powertrain can match or exceed the operational utility of a traditional racing engine without the high costs of professional EV systems. By using a 400V architecture, it enables compatibility with existing public charging networks, eliminating the need for specialized paddock infrastructure, a major barrier to amateur EV racing. Additionally, a 50,000 km service life proves that the ESU can operate across multiple racing seasons with minimal internal maintenance, something rare in EV motorsport.

Parameters:

Energy Capacity and Life Cycles

We do not use 100% of the battery's capacity. To ensure the car lasts for 50,000 km, we calculate the total energy (E_{total}) using this formula:

$$E_{total} = \frac{E_{model}}{DoD \cdot SoH} \quad (7)$$

Depth of Discharge (DoD): We limit the usable energy to avoid fully discharging the battery, which can shorten its lifespan [30]. By leaving a buffer (margin), we significantly extend its life.

State of Health (SoH): Batteries lose capacity as they age. This factor ensures that even when the battery is old after 50,000 km (or 1667 cycles), it still has enough energy to finish the race.

Voltage and Power

The system is designed around a 400V architecture. This specific voltage is chosen for three reasons:

- **Fast Charging:** It allows the use of standard road-side DC fast chargers, such as those in the Portuguese Mobi.E network, which provide DC output ranges compatible with 400V systems [31].
- **Inverter and Motor Range:** It stays within the range of our motor and inverter's accepted voltage (320-400 V) in all motor/inverter concepts.
- **Heat Reduction:** Higher voltage means we achieve the same power with less current. This reduces heat generated by Joule Effect.

Current Requirements:

Discharge ($I_{dis,peak}$): To hit 50 kW, we calculate the current needed when the battery is at its lowest voltage:

$$I_{dis,peak} = \frac{50,000 \text{ W}}{V_{min}} \quad (8)$$

Charge ($I_{ch,peak}$): The charging current is calculated as:

$$I_{ch,peak} = \frac{P_{avg} \cdot k_{peak} \cdot k_{loss}}{V_{sys}}$$

Where:

- $P_{avg} = 26 \text{ kW}$: Average charging power for 13kWh in 0.5 hours.
- $k_{peak} = 1.5$: Peak factor to account for 50% higher current during the initial constant current phase [32].
- $k_{loss} = 1.1$: Loss factor to include 10% overhead for cable, contactor, and internal resistance losses.

Selection and Technical Justification

In the concepts where the ESU is a battery, there are different cell options, we use a spreadsheet [35] algorithm to create a battery pack for each one. This allows us to see the pack's specifications before making a decision.

The tool takes the specs of a single cell and calculates the full pack using these steps:

Series Calculation (n_s): Stacking cells to reach the target voltage.

$$n_s = \frac{400 \text{ V}}{V_{cell,nom}} \quad (9)$$

Parallel Calculation (n_p): The number of parallel strings must satisfy three different limits: energy capacity, peak discharge current, and peak charge current. The algorithm calculates the minimum n_p for each and picks the largest value (max) to ensure the pack is never overstressed:

$$n_p = \max \left(\frac{E_{target}}{E_{cell} \cdot n_s}, \frac{I_{dis,peak}}{I_{cell,dis-max}}, \frac{I_{ch,peak}}{I_{cell,ch-max}} \right) \quad (10)$$

Comparing Cells

By plugging different cells into this spreadsheet, we can directly compare concepts while still meeting our 400V / 50 kW requirement. We used median values of cells that have the two most common chemistry types (LFP and NMC) and the three most common formats (Cylindrical, Pouch and Prismatic), as shown in Table 8:

Chemis.	Format	V_{nom}	Ah	$I_{ch}(A)$	$I_{dis}(A)$	$T_{max}(^{\circ}C)$	kg	€	Wh/L	$T_{ch}(^{\circ}C)$	Cycles	Data
LFP	Cylindrical	3.2	6	6	18	60	0.14	1.2	340	45	4000	[24]
LFP	Pouch	3.3	20	40	300	55	0.50	12	247	55	3200	[25]
LFP	Prismatic	3.2	107	107	320	65	2.0	35	385	65	4000	[26]
NMC	Cylindrical	3.6	5	25	100	80	0.07	5.3	670	65	1400	[27]
NMC	Pouch	3.3	39	555	740	55	1.0	9	450	55	2500	[28]
NMC	Prismatic	3.7	50	50	100	55	0.84	40	515	45	2500	[29]

Table 8: Reduced Comparison of Candidate Battery Cells for ESU Design

Concepts

The following four concepts are compared to determine the optimal ESU design, as shown in Table 9. Note that all concepts ensure a 50 kW peak power, and a 30km autonomy recovery in less than 30 minutes.

Concept	Detailed Description
Sustainable	25 kW Green H₂ Fuel Cell (FC) + Battery: A PEM FC stack utilizing green H ₂ . A small lithium battery helps reach the 50 kW peak, while the FC provides steady-state power.
Economical	LFP Battery Pack: 60% SoC (20–80%). Uses cheaper cells by restricting charging to nominal current, ensuring 50,000 km durability with lower lifecycle costs.
Performance	NMC Battery Pack: Optimized for maximum specific energy and power density. Operates within a 80% usable SoC window (10%–90%) to minimize vehicle mass while still providing sufficient thermal and degradation headroom for racing.
Innovative	Aluminum-Air FC + Battery: A "mechanical battery" exploring high theoretical energy density. Uses a consumable aluminum anode and alkaline electrolyte, coupled with a battery to help discharge in acceleration.

Table 9: ESU Conceptual Design Descriptions

Sustainability and Environmental Impact

LFP vs. NMC: LFP scores higher (3.5) due to its use of abundant, non-toxic materials (iron and phosphate), avoiding the environmental issues of cobalt and nickel mining in NMC (3.0). Its longer cycle life also reduces total material use over the vehicle's lifespan.

Green H₂ (Score 4): This high level of sustainability is only achievable at prototype scale. With a demand of < 2 kg per test session, 100% green hydrogen from renewable electrolysis can be sourced from low-volume suppliers, bypassing carbon-intensive "gray hydrogen." This makes maximum sustainability a unique feature of the prototype phase, which is not yet scalable, since it uses electricity that would otherwise be wasted. Furthermore, it poses a lower risk of supply chain disruptions compared to lithium batteries, which are largely sourced from China and may face challenges in the context of global geopolitical conflicts.

Aluminum-Air (Score 2): This concept is penalized due to anode consumption. Sustainability is hampered by the constant need to replace aluminum plates. While aluminum is recyclable, the industrial smelting process required to re-refine the used plates is exceptionally energy-intensive.

Technical Complexity Analysis:

Low Complexity (NMC/LFP – 4): Integration is primarily electronic, as it requires a Battery Management System (BMS) for safety. The solid-state nature makes mechanical integration straightforward.

High Complexity (H₂ Fuel Cell – 2): Beyond the stack, it requires a Balance of Plant (BoP): high-pressure storage (> 350 bar), secondary pressure regulators, an air delivery system (compressor/filter), and a humidification loop to maintain membrane health.

Highest Complexity (Al-Air – 1): This system requires a secondary electrolyte management system. In a racing environment, managing caustic circulation and preventing aluminum hydroxide buildup (clogging) adds significant mechanical risk compared to closed-cell batteries.

The following quantitative data justifies the the rest of the scores in the selection matrix, as summarized in Table 10:

Concept	Cost (€)	Efficiency (%)	Reliability MTBF (Hours)	Mass / Volume (kg / L)
Sustainable: Green H ₂ FC	10,000	50.0	3,500	130 kg / 140.0 L
Economical: LFP Battery	1,691	95.0	> 200,000	195 kg / 96.8 L
Performance: NMC Battery	5,342	92.5	> 200,000	71 kg / 54 L
Innovative: Al-Air FC	5,800	45.0	< 200	65 kg / 150 L

Table 10: Quantitative Concept Data

Trade-Off and Results

The trade summary table, showing normalized scores for each ESU concept, is presented in Table 11:

Metric	Sustainable	Economical	Performance	Innovative
Cost	1	5	3.5	3
Efficiency	2	5	4.5	1.5
Sustain.	4	3.5	3	2
Complexity	2	4	4	1
Perform.	3	1	5	3
Reliability	3	5	5	2

Table 11: Energy Storage Concept Trade-off Matrix

7.1.4 Cooling

Thermal management is a critical factor for the vehicle's reliability and performance. A standard 6-lap stint requires the powertrain and Energy Storage Unit (ESU) to be kept within their optimal operating windows without excessively penalizing the vehicle's overall mass or aerodynamic drag.

Market Overview

To properly position the Formula E-Vee in the current motorsport ecosystem, a benchmarking analysis was conducted against existing open-wheel racing categories.

- **Traditional Formula Vee (ICE):** The benchmark for low-cost racing. It relies on mechanically simple, passive/forced air and oil cooling via sidepod NACA ducts. While highly accessible, its carbon-emitting Internal Combustion Engine (ICE) technology is obsolete.
- **Electric Racing Karts:** Highly relevant for short-stint (10-15 minutes) electric racing. They successfully employ passive battery cooling - relying purely on the cells' thermal mass - and simple air-cooled motors, proving the viability of low-complexity EVs for sprint races.
- **Formula Student Electric (FSE):** University-level prototypes using advanced active thermal management (e.g., custom liquid cooling loops and 3D-printed micro-channels). While highly efficient for 80 kW limits, they are too complex, expensive, and fragile for sustainable amateur racing.
- **Formula E:** The pinnacle of EV racing, utilizing state-of-the-art technologies like direct immersion cooling. Its astronomical budgets and proprietary systems make it completely inaccessible to grassroots motorsport.

Concepts Selection and Technical Justification

Sustainable Concept - Green Liquid and Spray Cooling

This concept replaces traditional, highly toxic ethylene glycol with biodegradable heat transfer fluids, such as organic propylene glycol or deionized water with natural inhibitors - following the Organic Additive Technology (OAT) that makes it 'green coolant' [38]. Coupled with the high thermal mass of the Renault Zoe R110 motor, it provides a highly reliable, closed-loop liquid system with minimal environmental impact, and similar values of flow rate and heat absorption [39]. The usage of Fuel Cell also opens the possibility of the water produced by the reaction being sprayed onto the cells or radiators to drastically increase efficiency via 'Spray Cooling'.

Performance Concept: Dual Liquid and Forced Air Cooling

The motor and inverter share a high-efficiency liquid cooling loop featuring lightweight motorcycle radiators and SPAL high-static-pressure fans, ensuring the inverter remains below 85°C. Simultaneously, the battery pack is cooled independently via forced air routed through 3D-printed micro-channels. This setup is based on the higher power/mass ratio on the AMK motors [22], and several Formula Student prototypes, with the physical separation preventing the high-temperature motor coolant from thermally degrading the sensitive lithium-ion cells. It remains to be tested just how effectively air cooling for the battery can actually be implemented - by using a Simulink simulator that will be mentioned lower.

Economical Concept: Passive and Forced Air Cooling

Embracing the Formula Vee heritage of simplicity, this concept relies entirely on ambient air and structural thermal mass. By pairing an air-cooled inverter with a dense motor like the Bosch SMG 180 [20], the system simply absorbs the latent heat generated during the short 15-minute race stint without ever reaching the 130°C thermal cut-off limit. NACA (National Advisory Committee for Aeronautics) ducts integrated into the sidepods channel high-velocity air directly onto the inverter's fins and the battery pack's 3D-printed micro-channels. This approach eliminates the cost, weight, and mechanical complexity of pumps, radiators, and hoses, but has less performance and higher risk.

Innovative Concept: Phase Change Materials (PCM) and Heat Pipes

This concept explores zero-drag, solid-state cooling technologies. Subsystem components are wrapped in a graphene-infused wax matrix. As the vehicle races, the generated heat melts the wax, locking the battery and motor temperatures at the wax's exact melting point (e.g., 55°C) by exploiting the latent heat of fusion. Additionally, vacuum-sealed copper heat pipes are proposed to rapidly transfer localized hotspots to the vehicle's tubular steel chassis, effectively turning the entire frame into a giant passive heatsink [37].

Trade-Off and Results

Metric	Sustainable	Economical	Performance	Innovative
Cost	2	5	3	1
Efficiency	4	2	4	5
Sustain.	4.5	4	2	3
Complexity	3	5	2	2
Perform.	3	2	4	4
Reliability	3.5	5	3.5	4

Table 12: Cooling Concept Trade-off Matrix

From the Trade Summary Table 12, the Economical concept excels in cost, complexity and reliability due to the absence of moving parts, whereas the Performance concept is the only viable option if low-mass, high-density motors (like the AMK DD5) are selected. The Sustainable concept offers the most balanced approach.

7.2 Structures

7.2.1 Chassi

Market Overview

The selection of the vehicle's structural architecture was based on a comparative study of several established categories in motorsport. The analysis was based on competition vehicles with varying levels of complexity and performance, such as Formula Vee, Formula Student, and Sport Karts, where the balance between torsional rigidity, weight, and economic viability is a determining factor.

Market Options

Chassis Type	Advantages	Disadvantages	Cost
Steel space frame	Cheap, easy to manufacture	Heavier	Low
Aluminium space frame	Lightweight, corrosion resistant	Difficult to weld (TIG)	Medium
Hybrid (Steel + Composite)	Balanced weight and stiffness	Complex joints/bonding	High
Aluminium monocoque	Very light and rigid	Complex manufacturing	Medium-High
Carbon fibre monocoque	Extremely light and stiff	Extremely expensive	Very High

Table 13: Chassis Type Comparison including Hybrid and Aluminum Frame Options.

From a theoretical point of view, as the Table 13 shows, a steel tubular space frame chassis seems the most appropriate option due to its low cost and ease of manufacturing, although it is important to quantify it.

Manufacturing Processes and Material

The manufacturing processes required for the construction of the tubular chassis are relatively simple and well-established, beside of a composite one, involving techniques such as tube bending to shape the tubes, TIG welding to join frame elements, and laser cutting or CNC machining for precision parts.

Design Options

The chassis design options were developed according to the market analysis and future configurations based on the four main concepts defined by the group. The parameters created to evaluate which conceptual options align best with the project solutions are represented in Table 14.

Concept	Evaluation Parameter
Performance	Stiffness-Mass Ratio
Sustainability	Energy Manufacture Cost and Carbon Footprint
Innovation	"Out of the Box"
Economic	Cheaper

Table 14: Design Evaluation Parameters

Possible Concepts and Trade Summary Table

After the definition of all metrics and parameters, to evaluate various design configurations for chassis structure and fairing by giving a score between 0 and 5 to all concepts searched, in Table 16, corresponding to the properties of the material in Table 15, it is possible to have a structured Trade-off Analysis, designed to objectively compare various concepts with the high-level requirements of the project, with each metric.

Material	Density (g/cm^3)	Specific Strength ($MPa \cdot m^3/kg$)	Approx. Cost (€/kg)	Carbon Footprint ($kgCO_2eq/kg$)
Steel	7.85	60 – 90	1.5 – 3	2.0 – 3.0
Aluminium	2.70	90 – 105	5 – 9	12.0 – 16.0
Carbon Fiber	1.60	400 – 900	40 – 120	20.0 – 40.0
Fiberglass	2.10	150 – 250	10 – 25	7.0 – 10.0
Basalt Fiber	2.00	180 – 300	15 – 30	5.0 – 8.0
Kevlar	1.45	350 – 600	30 – 60	12.0 – 15.0
Plywood	0.65	50 – 80	2 – 5	-0.5 – 0.5
3DP PETG	1.27	30 – 50	15 – 25	4.0 – 6.0

Table 15: Material Properties

Example of an Tubular Chassi with Fairing and a Monocoque Chassi in Figure 8.

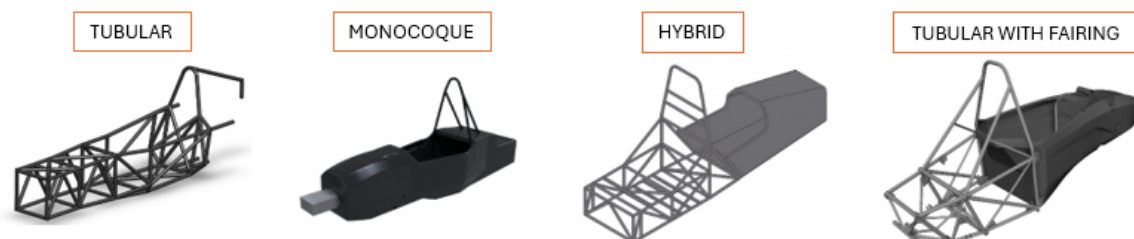


Figure 8: Concepts Options Description

ID	Concept	System	Efficiency	Performance	Cost	Complexity	Reliability	Sustainability
1.1	Sustainability	Structure - Tubular Recycled Aluminium	3	1	4	4	4	4,25
1.2	Economic	Structure - Tubular Steel	1	3	5	5	5	3,75
1.3	-	Structure - Aluminium Monocoque	4	4	2	3	4	1,25
1.4	Performance	Structure - Carbon Fiber Monocoque	5	5	1	3	3	1
1.5	Innovation	Structure - Hybrid (Steel & Carbon Fiber)	3	4	2	3	3	2
2.1	Economic	Fairing - Fiberglass	3	2	5	4	4	2,5
2.2	Performance	Fairing - Carbon Fiber	5	5	1	2	4	1
2.3	-	Fairing - Basalt Fiber	3	4	4	3	3	4
2.4	Sustainability	Fairing - PlyHood	2	2	4	4	2	5
2.5	Innovation	Fairing - 3D Printing	2	3	4	3	3	3,75
-	-	Metric	Mass (Kg)	Stiff/Mass (MPa · m3/kg)	Cost (€/Kg)	-	-	Energy Cost & Carbon footprint (kgCO2eq/kg)

Table 16: Chassi Concept Trade-off Matrix

Steel Tubular Structure (ID 1.2) emerged as the leading contender, primarily due to its superior behaviour in cost, complexity, and reliability metrics. In terms of fairing and bodywork, Basalt Fiber (ID 2.3) offers a peak sustainability rating and an excellent stiffness-mass ratio. Regarding aerodynamic devices, fairing would be the component to consider; however, the general objective is to minimize drag as much as possible, taking into account performance and energy consumption.

7.2.2 Suspension

The project requires a Volkswagen Beetle-based front axle, fixing the front suspension and steering geometry. This reduces cost, complexity, and technical risk through a proven system [5], but limits kinematic optimization and efficiency.

Design freedom is therefore concentrated on the rear suspension, which becomes the main subsystem for performance and integration improvements.

Market Overview

A market analysis was conducted to identify suspension layouts used in lightweight and performance vehicles. Trailing arm and semi-trailing arm systems are common in low-cost applications due to simplicity, robustness, and ease of manufacturing. In contrast, double wishbone systems are widely used in motorsport due to superior geometric control and tuning flexibility [40]. Hydraulic systems are typically limited to high-cost applications due to their complexity [41].

This comparison positions the project between simplicity and performance, requiring a suspension capable of supporting meaningful dynamic improvements.

Suspension Selection and Technical Justification

Four rear suspension concepts were evaluated based on cost, efficiency, sustainability, complexity, performance, and reliability.

The *trailing arm suspension* connects each wheel to the chassis through a pivoting arm. It is simple, robust, and low-cost, but provides limited camber and toe control [4].

The *semi-trailing arm suspension* introduces angled arms, enabling combined vertical and lateral motion. This improves camber behaviour compared to the trailing arm, at the expense of increased complexity.

The *double wishbone suspension* uses upper and lower control arms, allowing precise control of camber, roll center, and wheel path [40]. It offers superior performance but requires more components and complex integration.

The *hydraulic suspension* uses hydraulic elements to control wheel motion and, in some cases, ride height and load transfer [41]. Although highly capable, it is significantly more complex and costly.

Trade-Off and Results

A weighted trade-off matrix was used to compare the concepts (Table 17).

Metric	Semi-Trailing Arm	Trailing Arm	Double Wishbone	Hydraulic
Cost	4	5	2	1
Efficiency	3	2	5	4
Sustain.	4	4	3	2
Complexity	4	5	3	1
Perform.	3	2	5	4
Reliability	4	5	3	2

Table 17: Rear Suspension Concept Trade-off Matrix

7.2.3 Steering System

The steering system is constrained by the Beetle front axle, limiting optimization. The main objective is therefore proper integration, ensuring consistent response, reliability, and predictable handling.

Development focuses on compatibility, packaging, and robustness rather than alternative steering concepts.

7.2.4 Braking System

Regulations require all components of the wheel brakes to be VW Standard Type 1. Additionally, regulations require a dual-circuit braking system, with one circuit for the front wheels and another for the rear, both operated by a single pedal. This consists of two independent hydraulic systems, each with its own master cylinder and fluid reservoir. This design acts as a safety measure, ensuring that if one circuit fails, braking capability is still maintained through the other system.

Market Overview

VW Type 1 have most knowingly had drum brakes in both front and rear wheels. Later disc brakes were installed in front wheels. Nevertheless, no Beetle was ever fitted with rear disc brakes from factory.

Formula Vee cars typically use one of these three braking configurations: drum brakes on all four wheels, drum brakes at the rear with disc brakes at the front, or disc brakes on all four wheels (this last configuration requires for rear discs conversion kits or components from other VW models).

Braking System Selection and Technical Justification

Drum brakes are enclosed, protecting them from external debris, and are generally cheaper. They are widely used in Formula Vee due to regulatory acceptance. However, they dissipate heat less efficiently than disc brakes, and their more complex design can make maintenance more challenging. Disc brakes, by contrast, provide better cooling and more consistent braking performance, though their exposed design makes them more susceptible to dust, water, and wear.

From a sustainability perspective, drum brakes may be preferable. Their enclosed design retains a significant portion of wear particles, unlike disc brakes, which release particulate matter (PM) into the atmosphere. As brake wear is a major source of non-exhaust emissions, this reduces environmental and health impacts. A study on brake particle emissions ([43]) supports this approach, showing lower PM emissions for drum brake systems (Br4) compared to disc brakes (Figure 9). A comparison of advantages and disadvantages is presented in Table 18.

Disc Brakes		Drum Brakes	
Advantages	Disadvantages	Advantages	Disadvantages
Simple design	More expensive	Cheaper	Poor heat dissipation
Easy maintenance	Regulation constraints	Protected design	Complex maintenance
Good heat dissipation	—	Durable	Heavier
Lighter	—	—	—

Table 18: Brake System Type Comparison

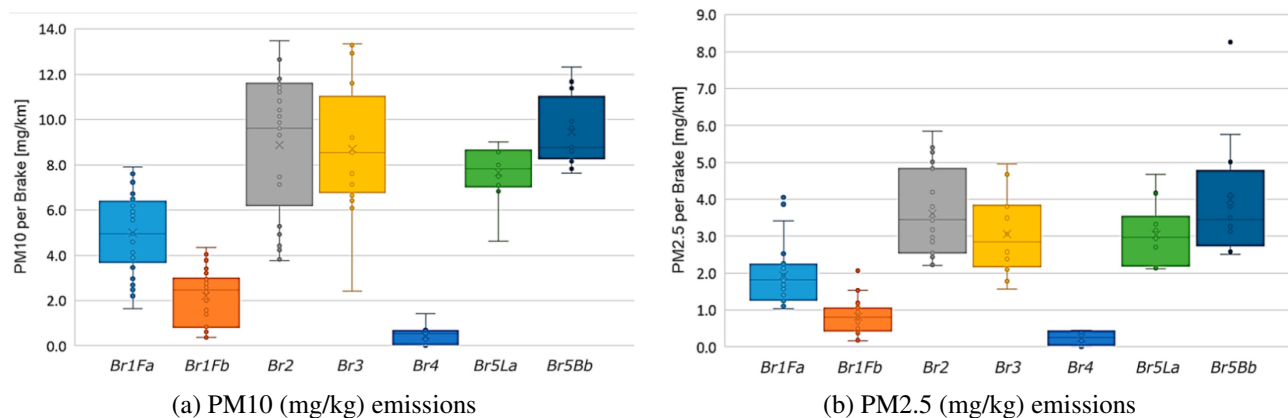


Figure 9: Particle Matter Mass Emissions

After an analysis of the differences between braking systems options, we can now determine the best option for each concepts (Table 19).

Sustainability	Economic	Performance	Innovative
Drum brakes front and rear	Drum brakes front and rear	Disc brakes (front) and drum brakes (rear)	Disc brakes front and rear

Table 19: Braking System Configurations by Concept

Trade-Off and Results

We can also assess them in a scoring matrix, according to the pre-established criteria (Table 20).

Metric	Sustainable	Economical	Performance	Innovative
Cost	4 (600–800 €)	4 (600–800 €)	3 (800–1k €)	1 (<< 1k €)
Efficiency	2	2	4	5
Sustain.	4	4	3	2
Complexity	4	4	3	2
Perform.	3	3	4	5
Reliability	3	3	4	4.5

Table 20: Braking System Concept Trade-off Matrix

7.2.5 Pedal Box

The pedal box will consists of two pedals: brake and throttle. The brake pedal is connected to a balance bar (allowing brake bias adjustments). For all concepts, the pedal box uses a simple housing, prioritising lightness and cost-effectiveness. The design must also consider cockpit space and integration within the chassis. Possible designs are shown below Figure 10.

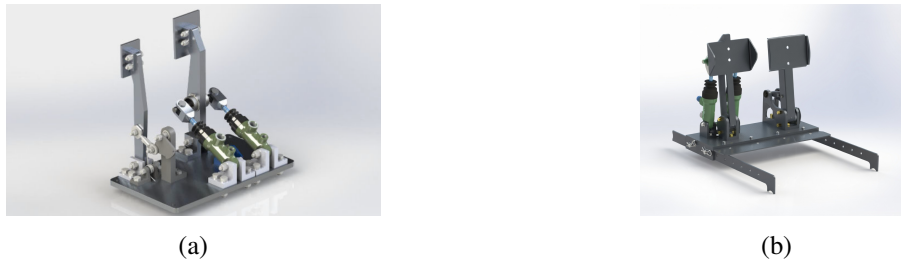


Figure 10: Possible Pedal Box Designs

8 Design Option Tree

The Design Option Tree (DOT) serves as a visual strategic map for the development of our project. It categorizes the vehicle into primary subsystems while mapping specific technological solutions to our four core pillars: Sustainability, Economy, Performance, and Innovation.

By employing this color-coded hierarchy (Figure 11), the DOT provides a clear overview of how each design choice aligns with our high-level requirements. This structured approach ensures that every subsystem offers a diverse range of concepts.

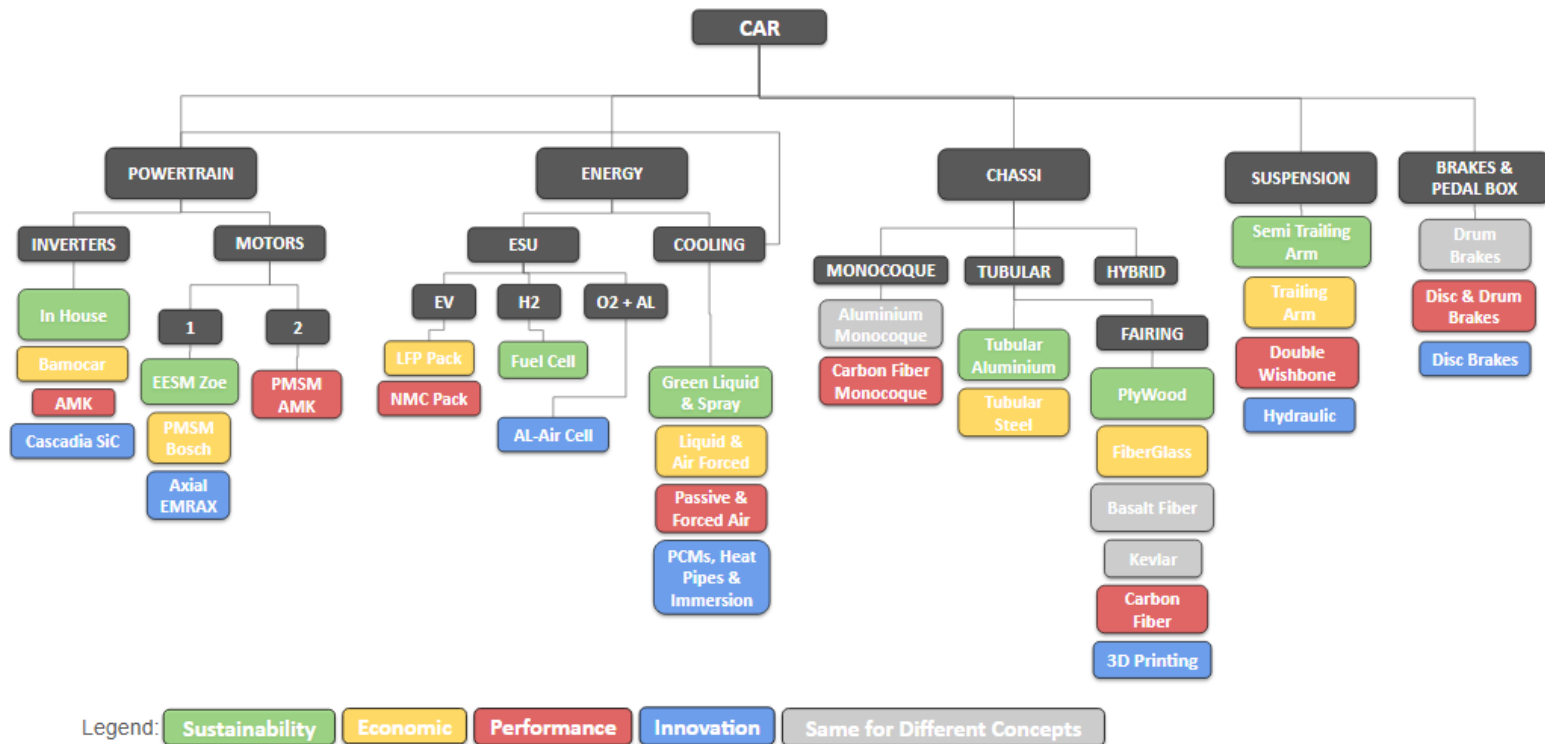


Figure 11: Design Option Tree

9 System Verification and Validation

At this midterm conceptual phase, the Verification and Validation (VV) process relies primarily on analytical models, datasheet cross-referencing, and theoretical feasibility studies to justify subsystem selection. Moving forward into the detailed design phase, the team will transition towards robust digital simulations and, ultimately, physical testing frameworks.

For the powertrain and thermal management systems (motors, inverters, and cooling), conceptual verification was achieved by cross-referencing OEM datasheets against preliminary Worst-Case Scenario (WCS) limits and first-principles thermodynamic calculations for heat generation. In the subsequent design phase, the inverter's electro-thermal map and a 1D thermal lumped-parameter cooling network (Simulink) will be integrated into the Lap Time Simulator (LTS) to confirm the system can sustain full race loads without thermal throttling. Additionally, Computer-Aided Design (CAD) packaging, FEA for motor mounts, and Computa-

tional Fluid Dynamics (CFD) for NACA duct optimization will ensure structural and aerodynamic compliance. Proposed physical validation includes dynamometer bench testing, high-voltage insulation checks, and fluid routing audits against FIA regulations.

Regarding the Energy Storage Unit, initial verification involved comparing cell specifications, analytical modeling for 26 kW charge times, and utilizing an n_p algorithm ($k_{peak} = 1.5$) for charge currents. Future physical validation would require timed 10–80% SoC charging sessions, DC-link logging, laboratory cell cycler testing for capacity and internal resistance, and hardware inspections over a simulated operational lifecycle.

Finally, chassis and suspension concepts were initially verified through literature review, weight estimation, baseline packaging models, and standard kinematic theory. Moving forward, a rigorous verification stage utilizing CAD and FEA will validate local and torsional stiffness, wheel travel, roll centers, and camber variation under representative dynamic load cases. If a prototype is manufactured, final validation will be conducted via test rig measurements and closed-course track testing to assess handling, stability, and overall structural safety.

10 Concept Selection Using the Analytic Hierarchy Process (AHP)

The Analytic Hierarchy Process (AHP) was used to select the most suitable electric Formula Vee concept among four alternatives: *Sustainable*, *Economical*, *Performance*, and *Innovative*. Each concept was evaluated across six criteria: cost, efficiency, sustainability, complexity, performance, and reliability, using weighted aggregation.

The evaluation was conducted at subsystem level (ESU, inverters, motors, cooling, chassis and aerodynamics, suspension and steering, and braking). Each subsystem was scored from 1 to 5 for each criterion, where 1 represents the worst performance and 5 the best; for cost, the scale reflects budget risk (5: minimal risk of exceeding the budget, 0: high risk); for efficiency, energy consumption (5: lowest consumption); for sustainability, environmental impact (5: lowest impact); for complexity, manufacturability and integration difficulty (5: simplest); for performance, dynamic capability (5: highest lap time potential); and for reliability, expected robustness and maintenance needs (5: highest reliability). Cost and complexity were computed using a simple average, while the remaining criteria were obtained through a weighted average, as defined in Equation (11):

$$Score = \sum (score_{subsystem} \times weight_{subsystem}) \quad (11)$$

The final score of each concept was then calculated as a weighted sum of all criteria (Equation (12)):

$$Final\ Score = \sum (score_{criterion} \times weight_{criterion}) \quad (12)$$

Criteria weights were defined based on the project *High-Level Requirements*. Cost (20%) ensures compliance with the 30,000€ budget constraint (OV-CT1). Efficiency (20%) and sustainability (20%) address the 30 km mission and zero-emission requirement (OV-OP4, OV-S1). Complexity (15%) and performance (15%) balance manufacturability with lap time targets (OV-C1, OV-OP5). Reliability (10%) ensures at least 100 km of daily operation (OV-OP6).

The AHP structure is shown in Figure 12, and subsystem weight distribution is presented in Table 21.

Criterion	ESU	Inverters	Motors	Cooling	Chassis & Aero	Susp. & Steering	Brakes
Efficiency	25%	15%	15%	10%	25%	5%	5%
Sustainability	20%	15%	20%	15%	20%	5%	5%
Performance	20%	15%	25%	10%	10%	10%	10%
Reliability	20%	20%	20%	10%	15%	5%	10%
Complexity	16.67%	16.67%	16.67%	16.67%	16.67%	16.67%	16.67%
Cost	16.67%	16.67%	16.67%	16.67%	16.67%	16.67%	16.67%

Table 21: Subsystem weighting for each evaluation criterion

The aggregated results are shown in Table 22.

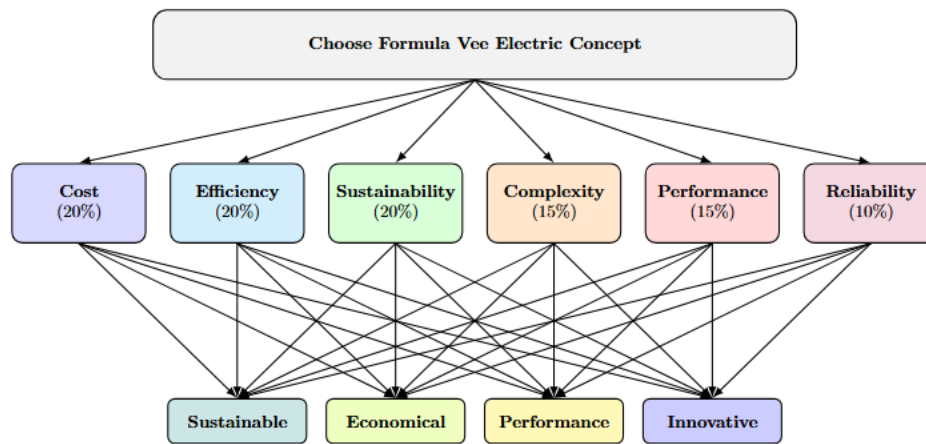


Figure 12: AHP structure for concept selection

Concept	Cost	Efficiency	Sustainability	Complexity	Performance	Reliability	Final Score
Sustainable	3.64	2.38	4.50	2.79	2.40	2.90	3.17
Economical	4.57	3.05	3.33	4.36	2.00	4.73	3.62
Performance	1.93	4.58	2.00	2.71	4.75	4.28	3.25
Innovative	1.86	3.28	2.68	1.79	3.85	3.10	2.72

Table 22: AHP Evaluation Results for each Vehicle Concept

The Economical concept achieved the highest score (3.62), offering the best balance between cost, efficiency, feasibility, and reliability. The Performance concept excelled dynamically but was penalized in cost and sustainability. The Sustainable concept performed well environmentally but lacked overall balance, while the Innovative concept showed higher complexity and risk.

Therefore, the Economical concept is selected for further development.

11 Sensitivity Analysis

To assess the robustness of the concept selection and understand how variations in priority affect the final ranking, a sensitivity analysis was performed. This involved varying the weights of the primary criteria (Figure 13; Figure 14) and adjusting the weights of the remaining criteria proportionally.

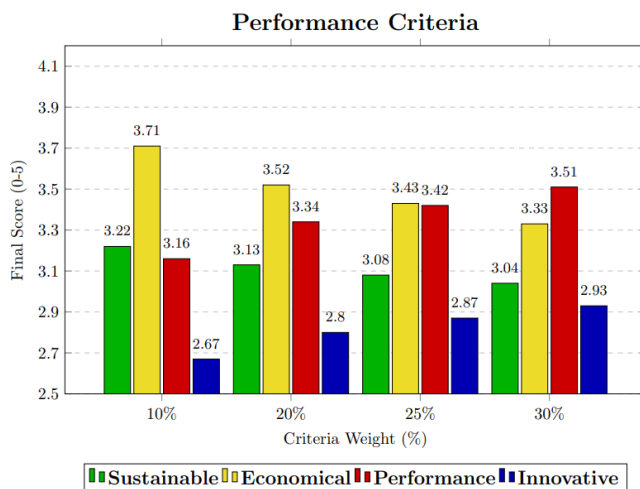


Figure 13: Sensitivity Analysis: Performance

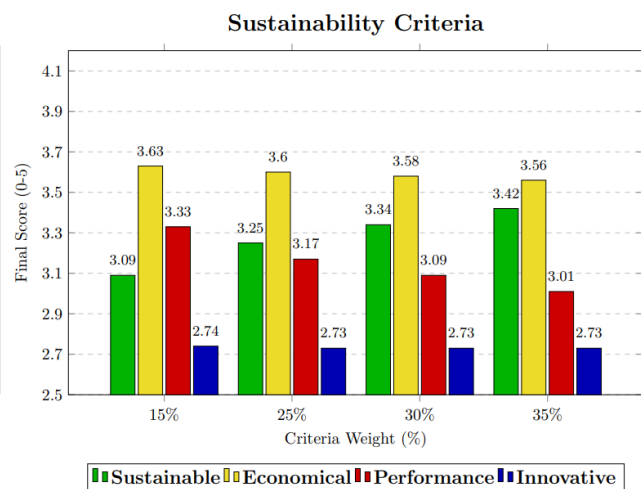


Figure 14: Sensitivity Analysis: Sustainability

The results indicate that the economical concept is consistently the top-performing option across most scenarios, while the performance concept emerges as a strong second-best overall solution, since it only appears as a leading concept when performance (and also efficiency) criterion is heavily prioritized.

In both sustainability and performance sensitivity tests, the economical concept achieves the highest scores even when these criteria are significantly increased in importance. The only exception is when the performance weight is around 30%.

The stability of the economical concept was observed across all simulations conducted for each criterion (omitted here for brevity). The study made supports its selection as the baseline concept for the Formula E-Vee prototype, as it proves to be the most balanced solution under varying project priorities.

12 Conclusion

The work presented in this Mid-Term Report marks the completion of the analytical and conceptual framework for the Formula E-Vee project. By deconstructing the vehicle into four specialized philosophies—Sustainable, Economical, Performance, and Innovative—it was possible to comprehensively map the current technological landscape against the project’s primary constraints. The transition from a qualitative market overview to structured trade-off matrices has provided a transparent and mathematically grounded justification for the path forward, ensuring that every compromise between cost, complexity, sustainability, and efficiency is carefully balanced.

The insights gathered from the benchmarking and detailed component evaluation have successfully narrowed the design space. Moving from a broad exploration of possibilities to a focused understanding of how each technical choice influences the vehicle’s overall dynamic behavior and environmental footprint, this systematic approach confirms the feasibility of a competitive electric Formula Vee. It demonstrates that the transition to a zero-emission powertrain can successfully uphold the category’s tradition of accessible racing while serving as a premium platform for engineering innovation.

Looking ahead, this report serves as the definitive guide for the upcoming Detailed Design phase. The immediate next steps will involve the convergence of these conceptual studies into the selection of a single, unified vehicle architecture, followed by the comprehensive engineering and dimensioning of the final prototype. The Validation and Verification procedures established throughout this document will act as the stringent benchmark against which the physical system is eventually measured. Ultimately, the robust foundation laid in this conceptual phase ensures that the final iteration of the Fittipaldi E-Vee will be a reliable, high-performance machine, ready to prove that grassroots motorsport can thrive in a green future.

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